Performance Work Statement (PWS)

Solid Waste Management

Fort Buchanan, PR

01 December 2025

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1.0 General Information.

1.1 Overview. Solid Waste Management is a comprehensive approach to managing non-hazardous solid waste that encompasses green procurement, waste prevention, and disposal programs. The Contractor, partnering with the installation, shall determine and utilize the most cost-effective Solid Waste Management service, emphasizing innovation and commercial best practices, to maximize diversion. The Contractor shall dispose of all commercial, industrial, and community activity waste in accordance with (IAW) all applicable Federal, State, and local laws and regulations. The waste shall be disposed of at a Subtitle "D" Municipal Solid Waste (MSW) landfill located in the Commonwealth of Puerto Rico and approved and fully permitted by the Department of Environmental Management. The Contractor shall pay all applicable fees associated with disposal.

1.2 Deliverables. The Contractor shall compile historical data, prepare reports, and submit information as specified by the Deliverables as directed by the Contracting Officer's Representative (COR) or Alternate Contracting Officer's Representative (ACOR).

1.3 Co-mingling Waste. The Contractor shall not co-mingle or transport MSW refuse with off-post generated MSW refuse. Solid waste (excluding regulated medical or hazardous waste) shall be collected, transported, and disposed of at the direction of the COR/ACOR.

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1.4 Outside Normal Duty Hours. Except for specials events, the Contractor shall be required to provide services during commercial hours, Monday through Saturday; excluding Federal holidays. Services outside of commercial hours must be authorized by the Contracting Officer (KO).

2.0 Description of Services.

2.1 Scope. The Contractor shall determine the best disposal method for solid waste management to avoid picking up an empty dumpster or container. The disposal method shall be approved by the KO. The Contractor shall empty refuse containers, compacters, and roll-offs no less than once a week. In the event of demand fluctuations, frequency adjustments must be vetted by the COR/ACOR and approved by KO. The Contractor shall remove all litter and waste on the ground within a ten (15) foot radius of the container. The Contractor shall track and report to the COR/ACOR all container weights and number of services per week on a monthly basis. The Contractor shall perform services IAW approved scheduled routes for pick-up and respond to calls for service when dumpsters reach full before scheduled pick-up.

2.1.1 Ordnance. Under no circumstances shall the Contractor knowingly accept ammunition, explosives, or firearms. The Contractor shall immediately report all potentially explosive items or materials to Directorate of Emergency Services (D.E.S.) emergency line and COR/ACOR.

2.1.2 Hazardous Materials. The Contractor shall not handle asbestos, human remains, or compact military food heaters. The Contractor shall not handle bio-medical (also known as “red bag”) or other hazardous materials. The Contractor shall report to the COR/ACOR any hazardous materials identified or discovered at the Collection Point.

2.1.3 Scavenging. The Contractor shall prohibit scavenging of materials from the Collection Point. The Contractor shall not salvage materials. The Contractor shall maintain the Collection Point to discourage the harboring, feeding, and breeding of disease vectors.

2.2 Schedules.

2.2.1 Refuse Collection Schedule. The Contractor shall coordinate with the COR/ACOR to develop a schedule that minimizes interference with Government business and submit the schedule to the KO for approval. The Contractor's schedule shall cover a twelve (12) month period. The Contractor shall submit the schedules not later than (NLT) ten (10) days prior to contract performance start date and every subsequent option period. The Contractor shall submit changes to the schedule to the KO NLT ten (10) working days prior to the proposed implementation date of any change. The Contractor shall not implement schedules or changes to the work schedule without prior written approval from the KO. The Contractor shall perform all work under the contract IAW the Contractor's schedules. All schedules shall be submitted to the KO for approval.

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2.2.2 Specific Tasks Route/Schedules Submittals. The Contractor shall submit a schedule organized in two (2) hour increments and by days of the week for vehicle routes and collection. The report shall be submitted for KO approval NLT ten (10) business days from the award date of the contract. The Contractor shall jointly monitor with the COR/ACOR the fill capacities during the first thirty (30) days of performance and one (1) other time during each performance period. Proposed changes to the schedule or haul route shall be approved by the KO in writing prior to route changes. Estimated container sizes and frequencies are found in Technical Exhibit 2. A map of the garrison’s current dumpster locations is shown in Technical Exhibit 3.

2.2.3 Weekly Refuse Work Schedule. The Contractor shall develop a weekly Refuse Work on Installation Schedule that includes, by service day and truck or route number, the driver, container locations in sequence of servicing, and the type of refuse container.

2.2.4 Changes to Pick-up Schedule. Frequency of servicing collection and refuse collection container size may be altered by the Contractor with approval from the KO to adjust to demand as determined by the Contractor during monitoring operations.

2.3 Recycle Support. N/A

2.4 Collection Containers.

2.4.1 Container Management. The Contractor shall place, combine, relocate, and remove all refuse containers and equipment in types and quantities, adequate to collect and deliver after receiving approval from the COR/ACOR. Damages to property or grounds after placing containers shall be repaired by the Contractor at no cost to the Government.

2.4.2 Containers and Installation Access. The Contractor shall place refuse containers according to historical locations or determine a better location for containers and propose the change to the COR/ACOR for approval. In addition, the refuse collection containers shall be positioned more than fifteen (15) yards from a food establishment's entrance doors and adjacent entry way, any and all exceptions, shall be approved by the Environmental Division.

2.4.3 Containers in Field Training Areas**.** The Contractor shall provide refuse containers to support training on ranges. The COR/ACOR will ensure all range and training area roads are accessible prior to collecting, disposing, or relocating containers.

2.4.4 HAZMAT in Containers. The Contractor shall notify the Directorate of Public Works (DPW) Environmental Division immediately after asbestos or other HAZMAT has been identified.

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2.4.5 Container Approval. All refuse collection containers (e.g., bins, totes, or cages) shall be KO approved. Containers must be clearly marked to identify names, telephone numbers, and emergency telephone numbers.

2.4.6 Serviceable Containers. The Contractor shall maintain refuse containers in a serviceable condition. The containers are unserviceable if any one of the following conditions exist:

a. Container does not meet specifications
b. Container is not leak proof (storm water pollution prevention compliant)
c. Container is not enclosed (except for roll-off containers designed to be open)
d. Doors, lids or hardware are missing, damaged, not operational, or do not close in the manner in which they were designed to close
e. Over twenty-five percent (25%) of exposed surface is not a uniform color (painted if container is not molded or composite plastic)
f. Over ten percent (10%) of exposed surface is covered with rust
g. Container exterior is damaged (substantial dents, holes, excessive scratches, burrs, ragged edges that present the container in an undesirable appearance).

2.4.7 Container Lids. All containers shall be equipped with metal lids and locking latches. All solid waste container tops, lids, and doors shall be closed after emptying.

2.4.8 Evaluating Containers. The Contractor shall evaluate the condition of collection refuse containers on a quarterly basis and provide this information in a report to the COR/ACOR. The report shall include container size, serial number, location, condition rating, number of times cleaned that quarter, number of times painted that quarter, the number of times repaired that quarter, number of dumpsters moved that quarter, total number of serviceable collection containers, and number of collection containers in need of repair.

2.4.9 Container Replacements. Deteriorated solid waste containers shall be replaced or repaired by the Contractor at no cost to the Government. The Contractor shall replace containers within two (2) working days of receiving notification. The Contractor shall furnish replacement containers when containers are removed for cleaning, painting, maintenance, or repair.

2.4.10 Container Spillage. The Contractor shall pick up all spillage resulting from emptying of container and compacting of all material waste and/or fluids produced by contractor vehicles.

2.4.11 Complaints. Complaints regarding unsanitary conditions due to Contractor negligence shall be corrected by the Contractor within two (2) hours of notification.

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2.4.12 Container Cleaning Schedule. The Contractor shall provide the COR/ACOR with a recommended refuse collection container cleaning schedule NLT ten (10) calendar days after contract award. The Contractor shall only clean refuse dumpsters containing food waste, (*e.g.*, containers at dining facilities, cafeterias, etc.) as needed. The Contractor shall provide the COR/ACOR with a list of refuse collection containers that have received cleaning service each month throughout the Period of Performance (POP). The Contractor shall provide a Monthly Refuse Container Cleaning Schedule that shall include days of service and driver, each container location, container type, cleaning time, and cleaning method. The Contractor shall clean and sanitize recycling/refuse containers as needed to ensure they are clean and odorless.

2.5 Collection Vehicles.

2.5.1 Vehicle Maintenance. Collection vehicles shall be maintained and serviced IAW the manufacturers' recommendations. The Contractor shall ensure collection vehicles receive periodic safety checks, and all irregularities shall be repaired before the vehicle is operated. The Contractor shall remove, all inoperable or unserviceable Contractor-furnished vehicles and equipment, off the installation. The Contractor shall not leave MSW in collection vehicles over twenty-four (24) hours. The collection vehicle operator shall immediately clean up all spills caused during the operation. All operators of Contractor-furnished vehicles shall be licensed by the State and operate IAW installation traffic regulations. The Contractor shall not leave vehicles unattended unless properly locked and secured in pre-approved location(s).

2.5.2 Unserviceable Vehicles. The Contractor shall have serviceable vehicles to ensure completion of contract requirements in the event of breakdowns, repair, or other mechanical problems. Vehicles that are identified as unserviceable shall be repaired or replaced NLT seven (7) calendar days at no additional cost to the Government. The Contractor shall provide maintenance and repair service.

2.5.3 Scales. Collection equipment of the types used for the collection, storage, and transportation of MSW shall meet the standards established by the American National Standards Institute, most current edition, established in ANSI Z245.1, (*Safety Standards for Refuse Collection Equipment*).

2.6 Special Events/Tasks.

2.6.1 Special Events Planning. The Contractor shall participate in the planning and coordinating of special events. Scheduled special events typically take place installation-wide.

2.6.2 Additional Services. The Contractor shall be required to provide an additional collections or relocate containers for special ceremonies, small construction projects, security, organizational events, unit moves, and to satisfy the Command. Relocations shall be accomplished, with the approval from the KO, not more than eight (8) working hours after the verbal request by phone. Major events requiring refuse and recycling support will be planned two (2) weeks in advance and the Contractor shall provide services necessary to support the event. The Contractor shall place refuse containers at the event or locations for the period of time specified by the COR/ACOR.

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2.6.3 Airfields. N/A

2.7 Other Events.

2.7.1 Storms and Natural Disasters**.** The Contractor shall provide additional collection containers or increase frequency of services to manage the temporary demand. The Contractor shall provide additional resources as needed to facilitate returning to full operational capacity.

2.7.2 Inclement Weather Planning. Upon notification of destructive weather by the COR/ACOR to prevent damage to containers, the Contractor shall collect all containers and temporarily relocate them to positions adjacent to buildings or other immovable structures. The Contractor shall provide appropriate tie-downs. The Contractor shall return all refuse collection containers to normal positions when destructive weather conditions have diminished.

2.7.3 Inclement Weather Delay. If the Contractor determines unsafe weather conditions for continued operations, then the Contractor shall request approval from the KO to suspend services. The Contractor shall suspend collection and disposal operations

because of inclement weather only after receiving approval from the KO. The Contractor shall resume operations with KO approval.

2.7.4 Unscheduled Clean-Ups. The Contractor shall remove materials on the ground at collection sites and clean-up unauthorized (illegal) dump sites. Examples may include abandoned property generated from unit moves and/or facility re-organization. The Contractor shall provide the appropriate containers requested.

2.7.5 Examples of Scheduled and Unscheduled Requirements. Scheduled requirements include the following: military unit moves, Armed Forces Day, military unit field exercises, Family Days, and Single Soldier Day, recovery operations due to natural causes or man-made causes, and military unit deployments. The Contractor shall provide collection containers to adequately manage the waste volume. Unscheduled events will be coordinated through the COR/ACOR. Upon notification, the Contractor shall provide the required services within five (5) working days.

2.7.6 Emergency Work. The KO may designate a work requirement as an emergency. The Contractor shall respond immediately to emergency work requests. The Contractor shall divert employees, if necessary, to respond to and accomplish emergency work. Emergency work would be necessary for the protection of life, health, or safety. Response time shall not exceed one (1) hour and work shall commence within two (2) hours upon notification. The Contractor shall continue working until the work is complete.

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2.7.7 Installation Closures. The Contractor shall have a Project Manager, designated as emergency essential, to handle refuse collections during installation closures IAW to DFARS 237.76.

2.8 Reports.

2.8.1 Monthly Summary Report. The Contractor shall provide a monthly summary report of tonnage landfilled as MSW and materials recycled to include, but not limited to, the following: electronics, bulk metal, paper, cardboard, co-mingled recyclables, Freon, tires, and organics. The monthly summary report shall, at a minimum, identify the alternative waste streams, C&D, organics, stable waste, landfill/transfer station tipping fees, and percent MSW and recyclable.

2.8.2 Work Control, Records, and Reports Plan (WCRRP). The Contractor shall develop and implement a WCRRP to include all information control procedures necessary to ensure timely and acceptable accomplishment of all contract data requirements; and permit tracking of work in progress. The Contractor shall submit a WCRRP to the KO for approval NLT ten (10) days after contract performance start date, and modifications to the plan NLT seven (7) working days prior to implementation of any changes. The Contractor shall revise, modify, and resubmit the plan for approval annually. The Contractor shall accurately, completely, and legibly prepare and submit the documentation and automated reports to the KO. Reports shall include any supplemental charts, photographs, manufacturer's pamphlets, etc., necessary to complete the report.

2.8.3 Container Condition Report. The Contractor shall evaluate the condition of the collection containers on a monthly basis and provide this information in a quarterly report to the COR/ACOR.

2.8.4 Weekly Skip Reports. The Contractor shall provide a Weekly Skip Report (WSR) to the COR/ACOR listing containers that cannot be serviced because of blocked access, contamination, illegal material, etc. The Contractor shall contact the COR/ACOR within (one) 1 hour after identifying the non-collected container. The Non-Collected Container Report shall contain the service date, truck or route number, driver's name, container location, time container was checked, why it could not be emptied, and comments (for notations such as a description and license number of blocking vehicle). Upon notification by the COR/ACOR, the Contractor shall ensure collection of the refuse collection container NLT four (4) hours following notification.

2.8.5 Regulatory Agency Inspections. The Contractor shall submit a written report of any inspection visit by representative(s) of any regulatory agency and immediately notify the KO. This report shall include the representatives' names, titles, identification numbers, and agencies, reason(s) for the visit, and comments or findings. The Contractor shall submit a copy of all reports received as a result of these visits to the COR/ACOR by the end of the next working day following receipt of the report. Any samples, reports, etc. shall be accompanied by a statement signed by the inspecting regulatory agency representative validating their authenticity.

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2.8.6 Other Reports. The Contractor shall provide copies of other reports to the COR/ACOR, such as those reports the Contractor develops to submit to State Regulators or to support the Contractor's work system, within one (1) working day of completion.

3.0 Government-Furnished Material (GFM)/Property (GFP)/Equipment (GFE).

3.1 Collection Stations. The DPW has designated collection stations. The Contractor may relocate or combine pick-up stations only after receiving approval from the KO.

3.2 Container Inventory. The Contractor shall conduct an inventory of all Government-owned recycle/refuse dumpsters, roll-offs, litter cans, compactors, and tippers. The inventory shall include the container type, size, condition, routine maintenance, repair or replacement date, repair type, location, buildings servicing, and inventory number. The inventory shall be provided within thirty (30) days of the start of each contract year.

3.3 Container Upkeep. The Contractor shall provide routine maintenance, repairs, and replacement to Government-owned vertical compactors, horizontal compactors, vertical balers, horizontal balers, dumpsters, litter cans, shredders, tippers, roll off containers, and roll away cover assemblies. The Contractor shall coordinate with the COR/ACOR all routine maintenance, repairs, and replacement.

3.4 Serialized Containers. Collection containers provided by the Government will have a number stenciled or painted on them for serialization and identification purposes.

3.5 Cubic Yard Capacity. Proposed capacities of containers based on historical data or estimated workload are indicated in Technical Exhibit 5. The Government reserves the right to change the cubic yard capacity at any of the collection sites, reduce or increase the frequency of a collection site, or change the number of collection sites as long as it does not increase the overall tonnage capacity requirement of the individual task order.

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4.0 Work Capability Levels.

TASK/SERVICE	CAPABILITY 1		CAPABILITY 2		CAPABILITY 3	
Routine Task/Service	**minimum level of service**	**Pricing**	**minimum level of service**	**Pricing**	**minimum level of service**	**Pricing**
Pickup - Refuse Dumpsters	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Pickup - Recycling Dumpsters	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Pickup - Food Waste Dumpsters	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Pickup - Organics (green/yard waste) Dumpsters	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Clean Refuse Dumpsters	**3Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**2Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Clean Food Waste Dumpsters	**3Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**2Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Maintain Dumpsters Appearance	**2Y** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1Y** - Patch holes, perform corrosion control, paint will match garrison color scheme, access doors are in good working condition, no visible crack on fork mounts	$	**1Y** - Patch holes, perform corrosion control, paint will match garrison color scheme, access doors are in good working condition, no visible crack on fork mounts	$
Pickup - MFH Garbage (non-RCI)	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Pickup - MFH Recycling (non-RCI)	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Pickup - MFH Green/Yard Waste	**2W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$	**1W** - sufficiently to inhibit the propagation or attraction of vectors and the creation of nuisances. (ref 1, 2,3)	$
Periodic Task/Service						
Unscheduled collection (storms, disaster, special events, etc)	Within 8 hours, prioritized as emergency SO (ref 2)	$	Within 12 hours, prioritized as emergency SO (ref 2)	$	Within **24 hours**, prioritized as emergency SO (ref 2)	$
Bulk Waste Item Pick –Up (units and non-RCI MFH)	**6Y**	$	**6Y**	$	**4Y** - 1x per quarter (ref 1)	$
Total Prices	**Total Price for Capability 1**	$	**Total Price for Capability 2**	$	**Total Price for Capability 3**	$

Legend:
A/R - As Required/Requested

D - Daily	1Y – 1 Time a Year
1W - Once a Week	2Y - 2 Times a Year
2W - Twice a Week	4Y - 4 Times a Year
3W - Times a week	5Y - 5 Times a Year
1M - Once a Month	6Y - 6 Times a Year
2M - Twice a Month	

Solid Waste Task Notes:

1. Deviations to the minimum level of service will require approval by the Command and the justification will take into consideration the collection cost, fuel consumption, generation rates, waste composition and storage capacity.

2. It is recommended that the costs associated with the collection and disposal of solid waste should be driven, completely or in part, by the quantity (weight) of waste generated and disposed/diverted. Avoid situations where the cost are based entirely on an annual fee or cost per container per pickup.

Cost Factor:

Potential Disposal Cost = Disposal Cost (w/out composting) + Program Cost + (Recycled waste in tons + Composted waste in tons) x Average Disposal Facility Fee (ADFF)

Actual Disposal Cost = Disposal cost + Cost for all programs – revenue from recycling

References:

1. 40 CFR 243.203
2. AR 420-1
3. TM 5-634

Fort Buchanan Site Map

UPDATED DATA ON SITE MAP DISTRIBUTION
SYSTEM REFLECTS CONDITIONS AS OF JUL 2019

Technical Exhibit TE-3: Refuse Special Events Collection Schedule
Installation Special Events

Special Activity	Description of Services	Qty of Dumpsters	Dumpster Capacity	Frequency
Armed Forces Day	Special Event	2	6CY	If Required
4th of July	Special Event	8	6CY	If Required
	Special Event	1	20CY	If Required
	Special Event	1	40CY	If Required
PRNG / USAR Centers	Special Event	8	6CY	If Required
Christmas Celebration	Special Event	4	6CY	If Required
Total Amount of Containers:		24		

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Technical Exhibit TE-2.1: Refuse Management Historical Data

Historical Refuse Collection Schedule

Installation Regular Service – 6 Cubic Yard Dumpsters

Bldg #	Location	# of Dumpsters	Dumpster Capacity	Pick Up Frequency	Total Yearly Pick Ups
21	Health Clinic	1	6CY	2 x Weekly	104
34,37,38	DPW O&M Shops	1	6CY	1 x Weekly	52
T-65	MWR Picnic Area	1	6CY	1 x Weekly	52
79	Navy Reserve	2	6CY	1 x Weekly	104
136	Morale Welfare & Recreation	1	6CY	1 x Weekly	52
148	MWR Community Services	1	6CY	1 x Weekly	52
151	Cyber Coffee	1	6CY	1 x Weekly	52
152	Welcome Center	1	6CY	1 x Weekly	52
159	Auto Craft Shop	1	6CY	1 x Weekly	52
161	Aqua Park (Water Spout)	1	6CY	2 x Weekly	104
167	Fitness Center	1	6CY	1 x Weekly	52
168	Bowling Center	1	6CY	3 x Weekly	156
172	[illegible]	1	6CY	3 x Weekly	156
181	Max[illegible]	1	6CY	1 x Weekly	52
192	CPAC	1	6CY	1 x Weekly	52
202	[illegible]	1	6CY	1 x Weekly	52
218	DES	1	6CY	1 x Weekly	52
224	USAR 2/348T	1	6CY	1 x Weekly	52
291	Kennel	1	6CY	1 x Weekly	52
292	Chapel	1	6CY	1 x Weekly	52
315	AFRC	2	6CY	1 x Weekly	104
357	USAR 1st MSC	2	6CY	1 x Weekly	104
369	Tennis Courts	1	6CY	1 x Weekly	52
390	HQ USAG	2	6CY	1 x Weekly	104
399	MICC/1St MSC	1	6CY	1 x Weekly	52
504	DOL	1	6CY	1 x Weekly	52
507	DOL	1	6CY	1 x Weekly	52
509	DOL	1	6CY	1 x Weekly	52
511	Training Center	1	6CY	1 x Weekly	52
512	SDDC	1	6CY	1 x Weekly	52
514	DHR/DPTMS	1	6CY	1 x Weekly	52
518	Post Library	1	6CY	1 x Weekly	52
522	1st MSC	1	6CY	2 x Weekly	104

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TE-2.1 Continued:

Bldg #	Location	# of Dumpsters	Dumpster Capacity	Pick Up Frequency	Yearly Pick Ups
523	1st MSC	1	6CY	1 x Weekly	52
527	NEC	1	6CY	1 x Weekly	52
528	1st MSC	2	6CY	1 x Weekly	104
538	Wash Rack	1	6CY	1 x Weekly	52
540	USFPO	1	6CY	1 x Weekly	52
541	USFPO	1	6CY	1 x Weekly	52
545	PRNG	3	6CY	1 x Weekly	156
TBD	PRANG – New Building	1	6CY	1 x Weekly	52
566	DDESS Admin Office	2	6CY	1 x Weekly	104
606	PXTRA	3	6CY	3 x Weekly	468
607	DPTMS/VI	1	6CY	1 x Weekly	52
608	MWR	1	6CY	1 x Weekly	52
609	Maintenance Shop	1	6CY	1 x Weekly	52
611	AAFES Military Clothing	1	6CY	1 x Weekly	52
612	AAFES WAREHOUSE	1	6CY	1 x Weekly	52
613	DOL/RMO	1	6CY	1 x Weekly	52
614	AAFES (Wareh[illegible])	1	6CY	1 x W[illegible]	52
615	Maintena[illegible]ce S[illegible]op	1	6[illegible]Y	1 x Weekl[illegible]	52
660	Commu[illegible]y C[illegible]b	2	6[illegible]	2 x Weekl[illegible]	208
673	Credit Unio[illegible]	1	6CY	1 x Weekl[illegible]	52
676	Veterinary Clinic	1	6CY	1 x Weekly	52
677	New Gas Station	1	6CY	1 x Weekly	52
689	Commissary	2	6CY	6 x Weekly	624
689	P1 Main Store	5	6CY	6 x Weekly	1560
1000	CDC	2	6CY	1 x Weekly	52
1100	Coconut Grove Playground	1	6CY	1 x Weekly	52
1140	Coconut Grove Skate Park	1	6CY	1 x Weekly	52
1300	BEQ	1	6CY	1 x Weekly	52
1301	BEQ	1	6CY	1 x Weekly	52
1307	BEQ	1	6CY	1 x Weekly	52
1318	1st MSC	2	6CY	1 x Weekly	104
Housing	Las Colinas Mailboxes	1	6CY	1 x Weekly	52
ACSS	High School (AHS)-	2	6CY	3 x Weekly	312
ACSS	Middle School	2	6CY	3 x Weekly	312
ACSS	Elementary School	3	6CY	3 x Weekly	468
Total Amount of Containers:		90	Total Amount of Yearly Pick Ups:		7,956

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Historical Refuse Collection Schedule

Installation Regular Service - Large Cubic Yard Dumpsters

Bldg/ Location	Description of Services	Qty of Dumpsters	Dumpster Capacity	Frequency	Yearly Pick Ups
37/DPW	Pick up debris -Wood & Vegetative	1	40CY	Bi-Weekly	52
	Pick up debris	1	20CY	Bi-Weekly	26
522/NG	Pick up debris	1	40CY	Monthly	12
600/DPW	Pick up debris	1	40CY	Bi-Weekly	26
613/AAFES	Pick up debris	1	40CY	Bi-Weekly	26
689/PX	Trash Compactor	1	36CY	3 x Week	156
689/AAFES	Trash Compactor	1	36CY	2 x Week	104
ACSS/DDESS	Pick up debris	1	20CY	Bi-Weekly	26
545/PRNG	Pick up debris	1	20CY	Monthly	12
541/PRNG	Pick up debris	1	20CY	Monthly	12
	Totals:		Total Amount of Yearly Pick Ups:		452

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Family Housing Quarters, Fort Buchanan, P.R.

Bldg Location	Description of Services	Qty of Dumpsters	Dumpster Capacity	Frequency	Yearly Pick Ups
Coconut Grove	Pick up Trash Cans	27	32 gals	1 x Weekly	1404
Las Colinas	Pick up Trash Cans	29	32 gals	1 x Weekly	1508
Total Amount of Containers:		56	Total Amount of Yearly Pick Ups:		2,912

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS
11/15

PART 1 GENERAL

APPLICABILITY: This entire new **Section 01 57 19** apply to all major construction, demolition, and/or renovation projects that will require both interior and exterior works at any Facility/Building in USAG Fort Buchanan PR.

For minor repairs to replace existing Buildings and/or Facilities, at least a General Permit for Other Works is required. When the project impacts more than 900 square meters (0.22 acres) and/or disturbs at least 40 cubic meters (52 cubic yards) of soil, and/or carry out activities in areas adjacent to bodies of water that require an Erosion Control and Sediment Control Plan (a "CES Plan") or a General Consolidated Permit, a Single Incidental Operational Permit (PUI) must be processed that consolidates four procedures: cutting and pruning authorization, a consolidated general permit, incidental activity and incidental activity for other projects or any other work that requires a General Permit and that is considered in Chapter 3.4 and 3.6 of the Joint Regulation of Permits (Joint Regulation 2020) and in the Regulation for the Processing of General Permits of the Department of Natural and Environmental Resources (PRDNER).

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1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846	Third Edition; (Update VII) Test Methods for Evaluating Solid Waste: Physical/Chemical Methods

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)FEDERAL REGISTER

29 CFR 1910.120	Hazardous Waste Operations and Emergency Response
32 CFR 651.4	Environmental Effects of Army Actions
33 CFR 328	Definitions
40 CFR 50	National Primary and Secondary Ambient Air Quality Standards
40 CFR 60	Standards of Performance for New Stationary Sources
40 CFR 61	National Emission Standards for Hazardous Air Pollutants

40 CFR 63	National Emission Standards for Hazardous Air Pollutants for Source Categories
40 CFR 64	Compliance Assurance Monitoring
40 CFR 68	Chemical Accident Prevention Provisions
40 CFR 112	Oil Pollution Prevention
40 CFR 122.26	Storm Water Discharges (Applicable to State NPDES Programs, see section 123.25)
40 CFR 152 - 186	Pesticide Programs
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 273	Standards for Universal Waste Management
40 CFR 273.2	Standards for Universal Waste Management - Batteries
40 CFR 273.3	Standards for Universal Waste Management - Pesticides
40 CFR 273.4	Standards for Universal Waste Management - Mercury Containing Equipment

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40 CFR 273.5	Standards for Universal Waste Management - Lamps
40 CFR 279	Standards for the Management of Used Oil
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 302	Designation, Reportable Quantities, and Notification
40 CFR 355	Emergency Planning and Notification
40 CFR 403	General Pretreatment Regulations for Existing and New Sources of Pollution
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions
49 CFR 171 - 178	Hazardous Materials Regulations

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EXECUTIVE ORDER

Executive Order 14057	Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability.
Executive Order 14008	Tackling the Climate Crisis at Home and Abroad, Section 103 (c), Section 201, Section 211
Executive Order 13990	Climate Crisis; Efforts to Protect Public Health and Environmental and Restore Science.
Executive Order 13834	Efficient Federal Operations is hereby revoked except for Sections 6. Duties of the Federal Chief Sustainability Officer, Section 7. Duties of Heads of Agencies, and Section 11. General Provisions

ARMY PUBLISHING DIRECTORATE

AR 200-1	Environmental Protection and Enhancement

U.S. DEPARTMENT OF DEFENSE-PUBLICATIONS

DOD DIRECTIVE 4715.21	Climate Change Adaptation and Resilience

DODD 4715.23	Integrated Recycling and Solid Waste Management. Establishes policy, assigns responsibility, and prescribes procedures to implement integrated solid waste management (ISWM) through waste prevention and recycling.
DoDI 4105.72	Procurement of Sustainable Goods and Services, September 7, 2016 (Change 1 August 31, 2018)
DoDI 4105.67	No appropriated Fund (NAF) Procurement Policy and Procedure, February 26, 2014 (Change 2 December 1, 2017)

DOD MEMORANDUM

Consideration of Greenhouse Gas Emissions and the Effects of Climate Change in Army National Environmental Policy Act Reviews, June 27, 2023

Hydrofluorocarbons and Their Alternatives for Mission Critical Military End Uses, December 15, 2022

United States Army Reserve Environmental Quality Policy

Environmental Justice Policy, January 19, 2021

Army Implementation Guidance for Electric Vehicle Charging Facility and Electric Vehicle Supply Equipment (EVCF/EVSE) Infrastructure, Charging Operations and Reimbursements

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Class Deviation-Prohibition on Procurement of Certain Items Containing Perfluorooctane Sulfonate or Perfluorooctanoic Acid, September 23, 2022

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(1996) U.S. Army Corps of Engineers Safety and Health Requirements Manual
WETLAND MANUAL	Corps of Engineers Wetlands Delineation Manual Technical Report Y-87-1

FORT BUCHANAN ENVIRONMENTAL POLICIES

POLICY MEMORANDUM NO. 8	Pollution Prevention Program
POLICY MEMORANDUM NO. 23-17	Fort Buchanan Environmental Sustainability
POLICY MEMORANDUM NO. 24	Conservation of Threatened and Endangered Species

STATE REQUIREMENTS, DRNA (Former PREQB) PERMITS AND REGULATIONS, FORT BUCHANAN REQUIREMENTS:

1. MEC pamphlet FY13 Oct.70 pub.
2. Sergeant Wolf 3Rs Poster (English)
3. Sergeant Wolf 3Rs Poster (Spanish)
4. Environmental Awareness Package (Documents DL#65-78)
5. SWMP-MS4 Construction SWPPP Template

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink. https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances#tab-1.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink. https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances#tab-2.

1.2.2 Contract Office Representative (COR)

Contracting officer's representative (COR-FAR Definition) means an individual, including a contracting officer's technical representative (COTR), designated, and authorized in writing by the contracting officer to perform specific technical or administrative functions.

1.2.3 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work but are not fully consumed during construction. Examples include, but are not limited to, excess paint thinner, methyl ethyl ketone, toluene, waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

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1.2.4 Environmental Pollution and Damage

Environmental pollution and damage are the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally, or historically.

1.2.5 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.6 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.7 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore

structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with 40 CFR 261.

1.2.8 Hazardous Materials

Hazardous materials as defined in 49 CFR 171 and listed in 49 CFR 172. Hazardous material is any material that: Is regulated as a hazardous material in accordance with 49 CFR 173; or requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.120; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D. Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this section for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.9 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibit a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D.

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1.2.10 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan.

1.2.11 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the "waters of the United States" must occur. Comply with federal, AR 200-1, state (Commonwealth of Puerto Rico), and local laws and regulations.

1.2.12 Land Resources IAW AR 200-1

Land resources are the ranges, cantonment areas, and associated natural resources to include soils and biota.

1.2.13 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by installations to obtain NPDES permit coverage for their storm water discharges.

1.2.14 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

1.2.15 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also mean materials, including wastewaters, centrifuge solids, filter residues or sludge, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with other state and local requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in other state regulations or local ordinances; the amount generated is "de minimums" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

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1.2.16 Pesticide

Pesticide is any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant.

1.2.17 Pesticide Treatment Plan “Integrated Pest Management”

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.18 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds, and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.19 Project Pesticide Coordinator “Installation Pest Management Coordinator”

The Project Pesticide Coordinator (PPC) is an individual who resides at a Civil Works Project office and who is responsible overseeing of pesticide application on project grounds.

1.2.20 Regulated Waste

Regulated waste are solid wastes that have specific additional federal, AR 200-1, state (Commonwealth of Puerto Rico), or local controls for handling, storage, or disposal.

1.2.21 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.22 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.22.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 2.5-inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may or may not be reinforced with or contain ferrous wire, rods, accessories, and welments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

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1.2.22.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.22.3 Material not regulated as solid waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.22.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 263.

1.2.22.5 Recyclables

Recyclables are materials, equipment, and assemblies such as, but not limited to, doors, windows, door, and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable. It also includes

commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable if sold to a scrap metal company. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

1.2.22.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.22.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.22.8 Wood

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Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.23 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, creeks or "waters of the United States". Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, AR 200-1, state (Commonwealth of Puerto Rico), and local laws and regulations.

1.2.24 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant.

1.2.25 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.26 Waters of the United States (WOTUS)

Until the Navigable Waters Protection Rule takes effect, the Step One rule is in effect. 40 CFR 230.3(s) indicates that the term “waters of the United

States" means the term as it is defined in Title 40 Chapter I Subchapter-D Part-120 Section 120.2.

a) Waters of the United States means:

1) Waters which are:

(i) Currently used, or were used in the past, or may be susceptible to use in interstate or foreign commerce, including all waters which are subject to the ebb and flow of the tide;

(ii) The territorial seas; or

(iii) Interstate waters;

2) Impoundments of waters otherwise defined as waters of the United States under this definition, other than impoundments of waters identified under paragraph (a)(5) of this section;

3) Tributaries of waters identified in paragraph (a)(1) or (2) of this section that are relatively permanent, standing or continuously flowing bodies of water;

4) Wetlands adjacent to the following waters:

(i) Waters identified in paragraph (a)(1) of this section; or

(ii) Relatively permanent, standing or continuously flowing bodies of water identified in paragraph (a)(1) or (a)(3) of this section and with a continuous surface connection to those waters.

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5) Intrastate lakes and ponds not identified in paragraphs (a)(1) through (4) of this section that are relatively permanent, standing or continuously flowing bodies of water with a continuous surface connection to the waters identified in paragraph (a)(1) or (a)(3) of this section.

Revised Definition of "Waters of the United States"
Waters of the United States

Definition of "Waters of the United States"
40 CFR PART 120 Revised Definition of ''Waters of the United States''; Conforming

1.2.27 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.28 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste

in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.3 GENERAL REQUIREMENTS:

The Contractor shall minimize environmental pollution and damage that may occur as the result of construction operations. The environmental resources within the project boundaries and those affected outside the limits of permanent work shall be protected during the entire duration of this contract. The Contractor shall comply with all applicable environmental Federal, AR 220-1, State (Commonwealth of Puerto Rico), and local laws and regulations. The Contractor shall be responsible for any delays resulting from failure to comply with environmental laws and regulations. The contractor is responsible for the preparation and submittal of all applicable PRDNER (Former PREQB) and OGPe (Single Business Portal – SBP) Permits.

The Contractor shall maintain permanent and temporary pollution control facilities and devices for the duration of the contract or for that length of time construction activities create the particular pollutant.

1.4 SUBCONTRACTORS:

The Contractor shall ensure compliance with this section by their subcontractors.

1.5 PAYMENT:

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No separate payment will be made for work covered under this section. The Contractor shall be responsible for payment of fees associated with environmental permits, application, and/or notifies obtained by the Contractor. All costs associated with this section shall be included in the contract price. The Contractor shall be responsible for payment of all fines/fees for violation or non-compliance with Federal, AR 200-1, State (Commonwealth of Puerto Rico), Regional and local laws and regulations.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government before is submitted to the PRDNER (Former PREQB) or the Single Business Portal (SBP) to obtain the required permits. Submittals with an "S" are for inclusion, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction/Demolition Submittals

Preconstruction Survey

NEPA Review as 32 CFR Part 651 [Environmental Analysis of Army Actions: Final Rule]

The National Environmental Policy Act 1969 requires that federally funded actions be evaluated for potential accumulative impact to the human environment, through the preparation of a Record of Environmental

Consideration (REC), Environmental Assessment (EA), or an Environmental Impact Statement (EIS).

As defined in 32 CFR Part 651, any Army structure may be a proponent. In general, the proponent is the unit, element, or organization that is responsible for initiating and/or carrying out the proposed action. The proponent is fully responsible for funding and preparation of the environmental documentation. This includes responsibility for the content, accuracy, quality, and conclusions of the NEPA analysis, even if another organization or a contractor prepares the resulting documentation.

The proponent, in coordination with the appropriate NEPA manager, must determine the appropriate level of NEPA analysis and documentation needed. The proponent must also identify the appropriate milestones, timelines, and inputs required for the successful integration of the NEPA process, including the use of scoping to further define the breadth and depth of required analyses. The proponent will assure completing the NEPA process prior to implementation of federally funded action; including those to be contracted out of government to private services.

Environmental Evaluation under Rule 141

PRDNER (Former PREQB)/OGPe (Single Incidental Operational Permit (PUI)) Environmental Permits:

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- Consolidated Permit; G
  - Solid Waste Management Permit (DS3
  - Control of Erosion and Sedimentation (CES) Pla
  - Dust Control

- General Permit for Other Works; G
  - CES Plan
  - Dust Control

- Air Emission Permit; G
  - Emission Source Permit
  - General Permit for emergency generators

- Asbestos Abatement Permit; G
  - Abatement Design Plan (Asbestos)

- Lead Abatement Permit; G
  - Abatement Design Plan (Lead)

- Solid Waste Management Permit; G

Regulatory Notifications; G

Environmental Protection Plan; G

US EPA Stormwater Notice of Intent (for NPDES coverage under the general permit for construction activities); G

SD-06 Test Reports; G

- Laboratory Analysis
- Inspection Reports
- Solid Waste Management Report; G

SD-07 Certificates

- Employee Training Records; G
- Environmental Manager Qualifications; G
- Certificate of Competency; G
- HAZMAT/HAZWOPER; G
- Lead/Asbestos Personnel Credentials; G

SD-11 Closeout Submittals

Stormwater Pollution Prevention Plan Compliance Notebook; G

Stormwater Notice of Termination (for NPDES permit for construction activities); G

Waste Determination Documentation; G

Disposal Documentation for Hazardous and Regulated Waste; G

Assembled Employee Training Records; G

Solid Waste Management Report, Permit, and Manifests; G

Hazardous Waste Manifests; G

Regulatory Closing Documents Notifications; G

Contractor Certification; G

Review for endorsement by the Environmental Division will be made for all required submittals in accordance with the requirements of the paragraph 1.12 ENVIRONMENTAL RECORDS BINDER.

1.7 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Prior to commencing construction activities or delivery of materials to the site, the Contractor shall submit an Environmental Protection Plan to the Contracting Officer. Review and approval shall be done by the DPW's (Directorate of Public Works) Environmental Division prior to be submitted to the Department of Natural and Environmental Resources, PRDNER (Former PREQB) or OGPE (SBP) for the required permits. The purpose of the Environmental Protection Plan is to present a comprehensive overview of known or potential environmental issues which the

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Contractor must address during construction. Issues of concern shall be defined within the Environmental Protection Plan as outlined in this section. The Contractor shall address each topic at a level of detail commensurate with the environmental issue and required construction task(s). Topics or issues which are not identified in this section, but which the Contractor considers necessary, shall be identified, and discussed after those items formally identified in this section. Prior to submittal of the Environmental Protection Plan, the Contractor shall meet with the Contracting Officer for the purpose of discussing the implementation of the initial Environmental Protection Plan; possible subsequent additions and revisions to the plan including any reporting requirements; and methods for administration of the Contractor's Environmental Plans. The Environmental Protection Plan shall be current and maintained onsite by the Contractor.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

1.7.1 Compliance

No requirement in this Section shall be construed as relieving the Contractor of any applicable Federal, AR 200-1, State (Commonwealth of Puerto Rico), and local environmental protection laws and regulations. During Construction, the Contractor shall be responsible for identifying, implementing, and submitting for approval any additional requirements to be included in the Environmental Protection Plan.

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1.7.2 Conformance with the Environmental Management System "Contractor requirement"

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract. All on-site contractor supervisory personnel shall become aware complete yearly DPW Environmental Division sponsored environmental training specified for the type of work conducted on-site. Upon inclusion in the contract Statement of Work, the Contracting Officer's Representative will verify that all contractor personnel have acquired the SEMS Program Awareness Package (Policy Letters, Posters and Brochures [Documents DL#65-78]) and posted at their appropriate site or location.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain

associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

The Contractor's personnel shall be trained in all phases of environmental protection and pollution control. The Contractor shall conduct environmental protection/pollution control meetings for all Contractor personnel prior to commencing construction activities and submit proof to the Contracting Officer.

1.8 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the following special environmental requirements and attached at the end of this document.

ENVIRONMENTAL PLANS

	DESCRIPTION	REVIEW AND ENDORSEMENT	APPROVING AUTHORITY
1	Environmental Protection Plan	Environmental Division, DPW, Fort Buchanan	Environmental Division, DPW, Fort Buchanan
2	Recycling & Solid Waste Minimization Plan	Environmental Division, DPW, Fort Buchanan	Environmental Division, DPW, Fort Buchanan
3	USEPA Notice of Intent for Storm Water Protection	Environmental Division, DPW, Fort Buchanan	Environmental Division, DPW, Fort Buchanan

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NOTE: The Contracting Officer for the proper evaluation and endorsement will forward Environmental Plans submitted by the Contractor to the Directorate of Public Works (PW) Environmental Division. Contractor shall allow 15 calendar days for review, endorsement and/or approval by the DPW Environmental Division.

ENVIRONMENTAL PERMITS

	DESCRIPTION	REVIEW AND ENDORSEMENT	APPROVING AUTHORITY
1	Categorical Exclusion Permit (Certification 4-C)	Environmental Division, DPW, Fort Buchanan	Puerto Rico Department of Natural and Environmental Resources, PRDNER (Former PREQB)/OGPe (SBP)
2	Consolidated Permit (Solid Waste, CES Plan, Dust Abatement Plan)	Environmental Division, DPW, Fort Buchanan	Puerto Rico Department of Natural and Environmental Resources, PRDNER (Former PREQB)/OGPe (SBP)
3	Other Permits, Considerations and Report (See Sections 1.10.3.5, 1.11, 1.12, 3.2, 3.6.2, 3.10 and 3.11)	Environmental Division, DPW, Fort Buchanan	By Applicable Authority

NOTE: The Application Forms for the Environmental Permits shall be evaluated and endorsed by the DPW, Environmental Division before the Contractor submit them to the PRDNER (Former PREQB) or OGPe (SBP), as applicable. Contractor shall allow 15 calendar days for review and endorsement by the DPW Environmental Division.

1.9 QUALITY ASSURANCE

1.9.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs, and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

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In accordance with the Endangered Species Act of 1973, Fort Buchanan is committed in the protection and conservation of the Puerto Rican Boa and the Palo de Rosa. Thus, all contractors must comply with Installation guidance and policies as specified in Section 1.10.3.5 Land Resources. Additional survey would be necessary if any endangered species is identified at the site work [Project Location].

1.9.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with Federal, AR 200-1, state (Commonwealth of Puerto Rico) and local regulations and as specified the Permit conditions. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. No work activities shall commence without submittal of any required regulatory notifications or permits to the Contracting Officer for DPW Environmental Division review. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.9.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP). Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

1.9.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating contractor compliance with Federal, AR 200-1, state (Commonwealth of Puerto Rico), local, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer.

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1.9.5 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with Federal, AR 200-1, State (Commonwealth of Puerto Rico) and local regulatory requirements for RCRA Very Small Quantity Generator. Provide a Position Description for each employee, by subcontractor, based on the Davis-Bacon Wage Rate designation or other equivalent method, evaluating the employee's association with hazardous and regulated wastes. Train personnel to meet Federal, AR 200-1 and State (Commonwealth of Puerto Rico) requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Contact additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States, and endangered species and their habitat that are known to be in the area. Provide copy of the Erosion and Sediment Control Inspector competence qualification.

1.9.5.1 Pest Control Program

Except as may be specified elsewhere in this contract, Contractor will not use or apply pesticides (such as herbicides or weed-killers, insecticides, or rodenticides) without the specific written prior approval of the Contracting Officer and the Installation Pest Management Coordinator (IPMC). Whenever a pest control "approved service" is contracted all the following will be considered:

a. The DPW Environmental Division will approve and monitor all DOD approved pesticide applications on the Installation and recommend modifications as needed.

b. All Pest Control Contracted services will provide evidence of certifications and training requirements for pesticide applicators and other personnel providing pest management support, such as pest surveillance and pest management. Contract oversight will be monitored by DPW Environmental Division Chief and the IPMC.

c. All Pest Control contracted services shall retain the following records: all pesticides application records, the list of pesticides and methods approved for use, the product labels and SDSs, the applicator names and certification documents, and appropriate contractor's Pesticide Business License(s) with categories of pest control, required training records and liability insurance policy certification. Maintain adequate records of pesticide disposal in accordance with Puerto Rico Environmental Law, EPA, and DOD procedures. Records will be provided as required.

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d. Termite treatments are mandatory prior to the construction of any new facility. The DPW Environmental Division must be informed the type of pesticide and the location/date/time of the application to assure proper pre-treatment.

e. All contractors will follow the specifications stated in the Fort Buchanan Integrated Pest Management Plan

Conduct a pest control meeting for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and pest infestation; familiarization with statutory and contractual pest control standards; installation and care of devices, and instruments, if required, for monitoring purposes to ensure adequate and continuous pest control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of waters of the United States, and endangered species and their habitat that are known to be in the area. Provide a Certificate of Competency for the personnel who will be conducting the pesticide application and management of pest control.

1.9.6 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with Federal, AR 200-1, State (Commonwealth of Puerto Rico) or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has

been taken. No time extensions will be granted, or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.10 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction/renovation process. This may include addressing the topics of air pollution, GHG emissions and climate change effects, asbestos/lead abatement, soil erosion and sediment control, stormwater pollution prevention, pest management, spill control, unexploded ordnance/munitions, explosive of concern, hazardous materials management, waste management, cultural and natural resources such as endangered species and tree management.

The Contractor will establish and implement an Environmental Protection Plan (EPP) by incorporating environmental related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required permits and reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 15 days after Contract award and not less than 10 days before the preconstruction meeting. Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or contract modifications that change the project scope of work in a way that could have an environmental impact. The requirement in this section will relieve the Contractor of any applicable Federal, AR 200-1, State (Commonwealth of Puerto Rico), and local environmental protection laws and regulations. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version onsite.

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The EPP includes, but is not limited to, the following elements:

1.10.1 General Overview and Purpose

1.10.1.1 Descriptions

Description of the Contractor's environmental protection personnel training program. A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as stormwater pollution prevention plan, including soil erosion and sediment control plan, spill control plan, solid waste management plan, wastewater management plan, air pollution control plan, contaminant prevention plan, preconstruction pest control and pesticide treatment plan, a historical, archaeological, cultural resources, biological resources such as endangered species and tree management and wetlands protection plan traffic control plan, Hazardous, Toxic and Radioactive Waste (HTRW) Plan, Non-Hazardous Solid Waste Disposal Plan and borrowing material plan, among other.

1.10.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

1.10.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

1.10.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.10.1.5 Contact Information

Emergency contact information (Name, office phone number, cell phone number, and e-mail address).

1.10.2 General Site Information

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1.10.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.10.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

1.10.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.10.3 Management of Natural Resources

1.10.3.1 Endangered Species

Currently, in Fort Buchanan are identified two federally designated endangered species: the Puerto Rican boa (Chilabothrus inornatus) and a small tree named Palo de Rosa (Ottoschulzia rhodoxylon). Also, the Puerto Rico

Department of Natural and Environmental Resources (PRDNER) considers those species as endangered and are protected under local regulations (Law Number 241 and Regulation Number 6766). In 2008, the U.S. Army Fort Buchanan (Fort Buchanan, Puerto Rico Department of Natural and Environmental Resources (PRDNER) and the U.S. Fish and Wildlife Service (USFWS) signed a Memorandum of Agreement (MOA) to protect and enhance the existing population of Palo de Rosa within Fort Buchanan. In 2015, a Memorandum of Understanding (MOU) between the Fort Buchanan, PRDNER and USFWS was replaced to include the management and conservation of the Puerto Rican boa (Chilabothrus inornatus). On September 2014, US Army Corps of Engineers prepared an Endangered Species Management 3 to guide the endangered species management program at Fort Buchanan.

a. Puerto Rican Boa Conservation Measures.

The U.S. Fish and Wildlife Service (Federal Register October 13, 1970) listed the Puerto Rican boa (Chilabothrus inornatus), formerly known as Epicrates inornatus, as endangered in 1970 and it is protected by the Endangered Species Act of 1973, as amended. Any person that injures, captures, or kills a Puerto Rican boa is subject to penalties under Federal law of up to $100,000, one year in prison or a combination of both. The U.S. Fish and Wildlife Service (hereafter the Service) has developed guidelines to avoid or minimize impacts on the boa during a project development in an area where the boa may occur. All contractors are required to follow these guidelines when working within or adjacent to boa habitat.

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1. A pre-construction meeting with DPW Environmental Division will be conducted to inform supervisors and employees about the conservation of protected species, as well as penalties for harassing or harming such species. Staff will be briefed on how to properly identify the boa and possible burrowing areas by the Environmental Conservation Manager. The contractor will receive boa awareness materials that should be always posted in areas were field staff/laborers can have access for viewing and self-orientation.

2. Prior to any earth movement or vegetation clearing, the boundaries of the project area, buffer areas and areas to be protected should be clearly marked in the project plan and in the field.

3. Prior to any use of machinery on areas where boa may occur, all personnel involved in site-clearing must be informed of potential presence of the snake, and the importance of protecting the snake. Safety Officer and or Project Manager shall continuously brief laborers of possible boa encounters and the importance of protecting the boa and the proper protocol to follow if sightings occur during construction process.

4. Before activities commence each workday during the vegetation clearing phase, the personnel experienced in identifying and searching for boas will survey the areas to be cleared that day. Research at Fort Buchanan has observed a higher encounter rate of boas in edge habitat in small and narrow forest fragments. Thus, clearings in or near forest perimeter/edge requires more search effort for boa surveys.

5. If boas are found within the working area, activities should stop at the area where the boas are found. Immediately call point of contact (POC) DPW Environmental Division, at telephone (787) 707-3575; 3508; 3522; 4863; 4864 to coordinate management actions. If relocation is deemed necessary, the snake should be relocated by authorized personnel that have completed the Boa handling training. A Puerto Rican Boa Sighting Form must be completed and submitted to DPW Environmental Division, 34 South Gate Road, USAG Fort Buchanan, PR 00934.

6. Strict measures will be established to minimize boa casualties by motor vehicles or other equipment. Before operating or moving equipment and vehicles in staging areas near potential boa habitats (within 10 meters of potential boa habitat), these should be thoroughly inspected to ensure that no boas are lodged in the standing equipment or vehicles. If boas are found within vehicles or equipment, authorized personnel must be notified immediately for proper handling and relocation. Any relocated boas should be transferred to appropriate habitat close to the project site.

b. Palo de Rosa
The Palo de Rosa is a small evergreen tree known only from Puerto Rico and Hispaniola (Dominican Republic and Haiti). The USFWS (Federal Register October 13, 1970) listed the Palo de Rosa (Ottoschulzia rhodoxylon) as an endangered species in 1990 and is protected by the Endangered Species Act of 1973(as amended). No critical Habitat has been designated for this species. The Recovery Plan for O. rhodoxylon was approved on September 20, 1994 (Service 1994). In Fort Buchanan are identified two separate populations of the Palo de Rosa located on the northeastern side of the haystack hill. Before commencing any new work or construction project at other designated area Palo de Rosa locations) contractor should consult with USFWS.

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1.10.3.2 Tree and Vegetation protection

a. Prevention of Landscape Defacement. Except in areas marked on the plans to be cleared, the General Contractor shall not deface, remove, cut, injure, or destroy trees or shrubs without specific written authority.

b. Trees designated to be saved shall be protected from either excavation or filling within the root zone. No ropes, cables, or guys shall be fastened or attached to any existing trees for anchorage unless specifically authorized by the contracting Officer. The General Contractor shall in any event be responsible for any damage resulting from such use.

1.10.3.3 Replacement of damaged landscape features

a. Restoration of Landscape Damage. Any trees or other landscape features scarred or damaged by the General Contractor's equipment or operations shall be restored as nearly as possible to the original condition at the General Contractor's expense. The Contracting Officer will decide what method of restoration shall be used, and whether damaged trees shall be treated and healed or removed and disposed of under requirements for clearing and rubbing. All scars made on trees not designated on the

plans to be removed by equipment construction operations, or by the removal of limbs larger than 1-inch in diameter shall be coated immediately with an approved tree wound dressing. All trimming or pruning shall be performed in an approved manner by experienced landscape personnel. A Tree Removal/Pruning Request Form must be completed and submitted to DPW Environmental Division, 34 South Gate Road, USAG Fort Buchanan, PR 00934.

b. Tree trimming with axes shall not be permitted. Trees that are to remain, either within or outside established clearing limits, that are subsequently damaged by the General Contractor and are beyond saving in the opinion of the Contracting Officer, shall be immediately removed, or replaced with a nursery-grown tree of the same species.

c. All new construction requiring tree removal, when deemed feasible within project design, shall include a mitigation plan by means of planting trees and/or other landscape vegetation. General Contractor shall submit to DPW Environmental Division a landscape drawing that adequately identifies location of specimens and species that will be used. DPW staff shall have a seven-work day period to review and for final approval of the plan. Priority shall be given to native species especially when pertaining to trees. New trees to be planted as mitigation for removed ones shall be at least six feet in height from the base of the root ball to the crown.

1.10.3.4 Temporary construction

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The Contractor shall, unless otherwise instructed in writing by the Contracting Officer, obliterate all signs of temporary construction facilities such as haul roads, work areas, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. The disturbed area shall be graded, filled and the entire area seeded unless otherwise indicated.

1.10.3.5 Wetland Areas

DOD's natural resources policy states that wetlands will be protected. All activities that affect wetlands require an environmental analysis in accordance with AR 200-1, 32CFR 651, and applicable Federal, AR 200-1 and state (Commonwealth of Puerto Rico) laws and regulations. USACE permits are required under Section 10 of the Rivers and Harbors Act of 1899 (RHA) prior to commencing any work or building any structures in navigable waters of the United States. Also, USACE permits are required under Section 404 of the Clean Water Act for the discharge of dredge or fill material into waters of the United States, including wetlands. The regulations established by Title 33 of the Code of Federal Regulations (CFR), Parts 320-330, prescribe the statutory authorities and general and special policies and procedures applicable to the review of applications for USACE permits.

Before commencing any new work in waters of the United States, a district engineer must be contacted, and a permit obtained, as appropriate (HQDA 1995). The USACE Antilles Regulatory Section is part of the Jacksonville District Regulatory Division and has regulatory responsibility for Puerto Rico and the US Virgin Islands.

Fort Buchanan has delineated areas considered as wetland inside perimeters to minimize impact and further degradation to this important habitat. Wetland areas delineated include a 12-acre forest fragment at the north side of the installation. In addition, two areas in the southern side (0.3 acres) of the installation have Swamp Bloodwood trees (Pterocarpus officinalis) which are considered obligate wetland vegetation. This tree species is of local concern and should be protected. For further details on management guidelines of wetland areas inside Fort Buchanan read the Wetland Delineation and Swamp Bloodwood Survey document. To acquire copy of the document contact Mr. Victor Rodriguez, Environmental Conservation Manager at 787-707-3508.

1.10.3.6 Land resources (General)

a. The Contractor shall confine all activities to areas defined by the drawings and specifications and shall provide effective protection for land and vegetation resources at all times.

b. General. It is intended that the land resources within the project boundaries and outside the limits of permanent work performed under this contract be preserved in their present condition or be restored to a condition after completion of construction that will appear to the natural and not detract from the appearance of the project. The General Contractor shall limit his construction activities to areas defined by the Drawings or Specifications.

1.10.4 Stormwater Management and Control

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a. Ground cover shall be provided with the established Best Management Practices (BMPs) to avoid erosion or exposed areas.

b. Erodible soils

1. The Puerto Rico Department of Natural and Environmental Resources, PRDNER(Former PREQB), Erosion Control and Sedimentation Prevention Regulation requires that all construction projects that impact more than 900 square meters (0.22 acres), and/or disturb at least 40 cubic meters, and/or conduct activities in area adjacent to water bodies, to submit an Erosion Control and Sediment Control Plan (a “Plan CES” in Spanish) to PRDNER for approval, as part of the General Consolidated Permit (Single Incidental Operational Permit (PUI)) or the General Permit for Other Work of the Office of Permits Management (OGPe) through the Single Business Portal (SBP).

2. Appropriate soil erosion and sediment controls must be used and maintained in effective operating condition during construction, and all exposed soil and other fills, as well as any work below the ordinary high-water mark, must be permanently stabilized at the earliest practicable date. Adequate sedimentation and erosion control management measures, practices, and devices, such as phased construction, vegetated filter strips, geotextile silt fences or other devices, shall be installed and properly maintained to reduce erosion and retain sediment on-site during and after construction.

c. Temporary measures

1. Structural Practices

2. Temporary and permanent stabilization

d. Effective selection, implementation, and maintenance of BMPs.

1. Under the Installation National Pollution Discharge Elimination System (NPDES) permit project must implement Best Management Practices (BMP), at a minimum, projects will require a Storm Water Pollution Prevention Plan (SWPPP) that addresses sediment and erosion control plan with appropriate BMPs.

2. BMPs shall be capable of preventing erosion, of collecting sediment, suspended and floating materials, and of filtering fine sediment, and shall be removed upon completion of work, and the disturbed areas stabilized. The sediment collected by these devices shall be removed and placed at an upland location, in a manner that will prevent its later erosion into a waterway or wetland. All exposed soil and other fills shall be permanently stabilized at the earliest practicable date.

1.10.5 Protection of the Environment from Waste Derived from Contractor Operations Control and disposal of solid and sanitary waste; and control and disposal of hazardous waste.

This item consists of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan. As a minimum, include the following:

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a. List of the types of hazardous wastes expected to be generated.

b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated.

c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications.

d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers).

e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted).

f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions (40 CFR 268).

g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar.

h. Used oil management procedures in accordance with 40 CFR 279; Hazardous waste minimization procedures.

i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.

1.10.6 Prevention of Releases to the Environment

Procedures to prevent releases to the environment. Notifications in the event of a release to the environment.

1.10.7 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. **The EPP will not be approved until the permits have been obtained.**

1.10.8 Clean Air Act Compliance

Equipment operation, activities, or processes performed by the Contractor shall be in accordance with all Federal, AR 220-1 and State (Commonwealth of Puerto Rico) air emission and performance laws and standards.

1.10.8.1 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may require Federal, AR 200-1, State (Commonwealth of Puerto Rico), or local permits under the Clean Air Act. Determine requirements based on any current installation permits and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager). Temporary Air Emission Permit require under Natural Resources and Environmental Department (former PREQB), Rule 203-204 of the Air Pollution Control Regulation. The application for a permit to operate a new or modified emission source shall be submitted to the Department of Natural and Environmental Resources at least sixty (60) days prior to the commencement of operations of a source.

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1.10.8.2 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used, or serviced. Comply with 40 CFR 60 Subpart IIII, 40 CFR 60 Subpart JJJJ, 40 CFR 63 Subpart ZZZZ, and local regulations as applicable. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation. Temporary Air Emission Permit will be required by the Natural Resources and Environmental Department (former PREQB), Rule 203-204 of the Air Pollution Control Regulation. The application for a permit to operate a new or modified emission source shall be submitted to the Department of Natural and Environmental Resources at least sixty (60) days prior to the commencement of operations of a source.

1.10.8.3 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants complies with 40 CFR 82 requirements. Technicians must be certified, maintain copies of certification on site, use certified equipment and log work that requires the addition or removal of refrigerant. Any refrigerant reclaimed is the

property of the Government, coordinate with the Installation Environmental Office and the DPW HVAC Shop to determine the appropriate turn in location.

1.10.8.4 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.10.8.5 Monitoring

For the protection of public health, monitor and control contaminant emissions to the air from Hazardous, Toxic, and Radioactive Waste remedial action area sources to minimize short-term risks that might be posed to the community during implementation of the remedial alternative in accordance with the following.

a. Perimeter Air Contaminant of Concern.

b. Time Averaged Perimeter Action Levels.

c. Perimeter Sampling/Monitoring Location[s].

d. Monitoring Instruments/Sampling and Analysis Methods.

e. Staffing.

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1.10.8.6 Compliant Materials

Provide the Government a list of MSDSs for all hazardous materials proposed for use on site. Materials must be compliant with all Clean Air Act regulations for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter, or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.11 LICENSES AND PERMITS

Contractor shall obtain licenses and permits required for the construction of the project and in accordance with Federal Acquisition Regulation FAR 52.236-7 Permits and Responsibilities. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7 Permits and Responsibilities. The contractor is responsible for the preparation and submittal of all applicable Federal, Army 200-1, and State Permits (Commonwealth of Puerto Rico).

1.11.1 Permits for Equipment Used by Contractor in Performing Contracts.

For equipment used in the performance of this contract, Contractor shall obtain in Contractor's name and at no additional expense, all permits, coordination, certifications, or other regulatory authorization necessary to perform and complete the work required by this contract under applicable environmental laws and regulations. “Applicable environmental laws and regulations" includes but is not limited to: the Clean Air Act, Clean Water Act, Resource Conservation and Recovery Act, Comprehensive Environmental

Response, Compensation and Liability Act (CERCLA), Toxic Substances Control Act, Federal Insecticide Fungicide and Rodenticide Act, Coastal Zone Management Act, Endangered Species Act, National Historic Preservation Act, Safe Drinking Water Act, Emergency Planning and Community Right-to-Know Act, Oil Pollution Act, and State, County, and Local laws and regulations on the same subjects.

1.11.2 Permits Needed for Construction, Excavation, Modification, Renovation, Demolition, Installation, or Other Alteration of Buildings, Structures, Equipment, Installations, Real Property or Systems.

Contractor shall identify all Federal, Army 200-1, State (Commonwealth of Puerto Rico), County, or local, permits, coordination, certifications or other regulatory authorization requirements under all applicable environmental laws and regulations as defined in (a.) above. Contractor shall then prepare and submit in draft all applicable permit applications, coordination, notices, or other required filings, together with all supporting data to the contracting officer for review. Permit applications or notifications or other documents that must be submitted by DPW will be submitted by DPW, and any documents that must be submitted by the contractor will be returned after review to the contractor for submission. No work requiring permit or other written authorization shall proceed before the Contractor has the permit or authorization or a copy.

1.12 ENVIRONMENTAL RECORDS BINDER

Contractor shall maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

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1.13 PESTICIDE DELIVERY, STORAGE, AND HANDLING

1.13.1 Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels indicating the EPA registration number and the manufacturer's registered uses. Store pesticides according to manufacturer's instructions and under lock and key when unattended. When applicable, all pesticide delivery, storage or handling must be previously approved by the Installation Pest Management Coordinator, DPW Environmental Division.

1.13.2 Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during each phase of the application. Furnish SDSs for pesticide products.

1.14 SOLID WASTE MANAGEMENT PERMIT

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or license

showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.14.1 Solid Waste Management Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

1.15 FACILITY HAZARDOUS WASTE GENERATOR STATUS

Buchanan is designated as a Very Small Quantity Generator (VSQG). Contractor shall meet the regulatory requirements of this generator designation for any work conducted within the boundaries of this Installation. Comply with provisions of Federal, AR 200-1, State, and Local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes.

PART 2 PRODUCTS

Products and materials used in this contract shall be, to the greatest extent practicable and consistent with financial prudence, efficient (Energy Star or Federal Energy Management Program (FEMP) designated), water-efficient (to meet EPA criteria such as Water Sense), bio based, environmentally preferable (e.g., Electronic Product Environmental Assessment Tool (EPEAT) certified), non-ozone depleting, contain recycled content, or are non-toxic or less toxic alternatives, where such products meet agency performance requirements, as per Executive Orders (EOs) 14057 and 13834. The Contractor shall comply with the following requirements:

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a. Utilize performance contracting to achieve energy, water, building modernization, and infrastructure goals.
b. Ensure that new construction and major renovations conform to applicable building energy efficiency requirements and sustainable design principles; consider building efficiency when renewing or entering into leases; implement space utilization and optimization practices; and annually assess and report on building conformance to sustainability metrics.
c. Implement waste prevention and recycling measures and comply with all Federal requirements regarding solid, hazardous, and toxic waste management and disposal.
d. Acquire, use, and dispose of products and services, including electronics, in accordance with statutory mandates for purchasing preference, Federal Acquisition Regulation requirements, and other applicable Federal procurement policies.

2.1 GREEN PROCUREMENT:

To comply with the green procurement requirements of Section 6002 of the Resource Conservation and Recovery Act of 1976 (RCRA) and Executive Order, the Government requires the use of recycled, bio-based, biodegradable, non-hazardous, and/or recovered materials and products identified in the Environmental Protection Agency's (EPA) Comprehensive Procurement Guidelines. These materials and products must meet the requirements of the PWS, must not delay the progress of work, must be available from more than one source, and must not be cost prohibitive. EPA guideline items are seen as the minimum

that should be considered when evaluating recyclable materials. Other materials and products not listed, but commonly used in industry outside of the Government, should also be considered. Material and product submittals for all recycled-content items should list the recycled and recovered materials used and the percentage content. The Contractor shall track the purchase, quantities, and the percent of recyclable content for these materials throughout the life of the contract. The Contractor shall consolidate and provide this information to the COR monthly in reports. The Contractor shall annually submit a summary report of certification and estimate as required by FAR 52.223-9, title, Certification and Estimate of Percentage of Recovery Material Content for EPA Designated Items, upon completion of the base year and each follow-on year, until completion of the contract to the COR. The intended end use of the item, materials with recovered materials content, at levels recommended by the EPA's CPG is found at website; https://www.epa.gov/smm/comprehensive-procurement-guideline-cpg-program.

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

The Contractor shall minimize interference with, disturbance to, and damage to natural resources including fish, wildlife, plants, and their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility. The following endangered species are known and could be affected within the Installation limits: Puerto Rican boa (Chilabothrus inornatus) and Palo de Rosa (Ottoschulzia rhodoxylon).

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Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified. Contractor shall ensure training and distribution of educational material to all personnel.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting

Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.1.3 Streams

Stream crossings must allow movement of materials or equipment without violating water pollution control standards of the Federal, AR 200-1, State (Commonwealth of Puerto Rico), and Local governments. Construction of stream crossing structures must be in compliance with any required permits including, but not limited to, Clean Water Act Section 404, and Section 401 Water Quality.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work and repair the area to its original condition unless otherwise required by the Contracting Officer.

3.2 STORMWATER

Do not discharge stormwater from construction sites to the sanitary sewer. If the water is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

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3.2.1 Construction General Permit

Contractor shall provide a Construction General Permit as required by 40 CFR 122.26, EPA and any other applicable regulation and permits from the Commonwealth of Puerto Rico. Under the terms and conditions of the permit, install, inspect, maintain BMPs, prepare stormwater erosion and sediment control inspection reports, and submit SWPPP inspection reports. Maintain construction operations and management in compliance with the terms and conditions of the general permit for stormwater discharges from construction activities.

3.2.1.1 Stormwater Pollution Prevention Plan

Contractor shall submit a project-specific Stormwater Pollution Prevention Plan (SWPPP) to the Contracting Officer for approval, prior to the commencement of work. The SWPPP must meet the requirements of 40 CFR 122.26, EPA General Permit, and the Commonwealth of Puerto Rico General Permit for stormwater discharges from construction sites.

Include the following:

a. Comply with terms of the Federal, Army 200-1 and State (Commonwealth of Puerto Rico) general permit for stormwater discharges from construction activities. Prepare SWPPP in accordance with the Federal and/or state (Commonwealth of Puerto Rico) requirements. Use Federal guide Developing your Stormwater Pollution Prevention Plan located at http://water.epa.gov/polwaste/npdes/stormwater/Stormwater-Pollution-Prevention-Plans-for-Construction-Activities.cfm to prepare the SWPPP.

b. Select applicable BMPs from EPA Fact Sheets located at http://water.epa.gov/polwaste/npdes/swbmp/Construction-Site-StormWater-Run-Off-Control.cfm or in accordance with applicable state or local requirements.

c. Include a completed copy of the Notice of Intent, BMP Inspection Report Template, and Stormwater Notice of Termination, except for the effective date.

3.2.1.2 Stormwater Notice of Intent (NOI) for Construction Activities

Prepare and submit the Notice of Intent for NPDES coverage under the general permit for construction activities to the Contracting Officer for review and approval.

Prepare and submit a Notice of Intent as a co-permittee to the Contracting Officer, for review and approval. Submit the approved NOI and appropriate permit fees onto the appropriate Federal or state agency for approval. No land disturbing activities may commence without permit coverage. Maintain an approved copy of the SWPPP at the onsite construction office, and continually update as regulations require, reflecting current site conditions.

3.2.1.3 Inspection Reports

Submit "Inspection Reports" to the Contracting Officer in accordance with Federal and State (Commonwealth of Puerto Rico) of Construction General Permit.

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3.2.1.4 Stormwater Pollution Prevention Plan Compliance Notebook

Create and maintain a three-ring binder of documents that demonstrate compliance with the Construction General Permit. Include a copy of the permit Notice of Intent, proof of permit fee payment, SWPPP and SWPPP update amendments, inspection reports and related corrective action records, copies of correspondence with the Federal, Army 200-1 and State (Commonwealth of Puerto Rico) Permitting Agency, and a copy of the permit Notice of Termination in the binder. At project completion, the notebook becomes property of the Government. Provide the compliance notebook to the Contracting Officer.

3.2.1.5 Stormwater Notice of Termination (NOT) for Construction Activities

Submit a Notice of Termination to the Contracting Officer for approval once construction is complete and final stabilization has been achieved on all portions of the site for which the permittee is responsible. Once approved, submit the Notice of Termination to the appropriate State or Federal agency. Prepare as-built topographic survey information required by the permitting agency for certification of the stormwater management system and provide to the Contracting Officer.

3.2.2 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance Federal, State local laws and regulations and Installation Policies or BMP’s. Preserve vegetation to the maximum extent practicable. Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

3.2.2.1 Erosion Control

Prevent erosion by mulching, Compost Blankets, Geotextiles, temporary slope drains. Stabilize slopes by sodding or such combination of these methods necessary for effective erosion control.

Provide seeding in accordance with UFGS Section 32 92 19 SEEDING.

3.2.2.2 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Include the following devices: silt fence, temporary diversion dikes, storm drain inlet protection. Location and details of installation and construction are indicated on the drawings.]

3.2.3 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

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3.2.4 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2.5 Municipal Separate Storm Sewer System (MS4) Management

Comply with the Installation's MS4 permit requirements.

3.3 SURFACE AND GROUNDWATER

The Contractor shall monitor construction activities to prevent pollution of surface and ground waters. Toxic or hazardous chemicals shall not be applied to soil or vegetation unless otherwise indicated. All water areas affected by construction activities shall be monitored by the Contractor.

3.3.1 General

The General Contractor shall not pollute storm drainage, streams, lakes, or reservoirs with fuels, oils, bitumens, calcium chloride, acids, construction wastes or other harmful materials or pollutants. It is the responsibility of the Contractor to determine and comply with all applicable Federal, AR 200-1,

State (Commonwealth of Puerto Rico), regional, municipal, and other regulations.

3.3.2 Spillage

The Contractor shall take special measures to prevent chemicals, fuels, oils, greases, bituminous materials, waste washings, herbicides, cement, and surface drainage from entering public waters. In the event of a spill, the contractor must make all required notifications to Federal, State (Commonwealth of Puerto Rico) or Local authorities and will notify the Contracting Officer and the On-Scene Coordinator immediately. All spills must be reported to the Fort Buchanan Fire Department Office at 4911, 5911, 787-707-3410 or 3337; and to the DPW Environmental Division at 787-707-3575 / 3522 / 3572.

3.3.3 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing state water quality standards and designated uses of the surface water body. Comply with State (Commonwealth of Puerto Rico) water quality standards (PREQB Regulation) and anti-degradation provisions and the Clean Water Act Section 404.Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

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If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer and the Federal or State authority, as applicable. Discharge of hazardous substances will not be permitted under any circumstances.

3.3.4 Waters of the United States

Do not enter, disturb, destroy, or allow discharge of contaminants into waters of the United States, except as authorized herein. The protection of waters of the United States shown on the drawings in accordance with paragraph LICENSES AND PERMITS is the Contractor's responsibility. Authorization to enter specific waters of the United States identified does not relieve the Contractor from any obligation to protect other waters of the United States within, adjacent to, or in the vicinity of the construction site and associated boundaries.]

3.4 PROTECTION OF CULTURAL RESOURCES

3.4.1 Archaeological Resources

At the Federal level, cultural resources are protected by the National Historic Preservation Act (NHPA; Public Law 89-665) of 1966, as amended, which was enacted to preserve historical and archeological sites in the U.S. Regulations for federal projects. Existing archaeological resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract. If,

during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include but are not limited to any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Contracting Officer so that the appropriate authorities may be notified, and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

3.4.2 Historical Resources

Section 106 of the NHPA requires federal agencies to consider the effects of their undertakings on historic properties and afford the Advisory Council on Historic Preservation (ACHP) an opportunity to comment on such undertakings. Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

3.5 AIR RESOURCES

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Equipment operation, activities, or processes will be in accordance with 40 CFR 64 and state air emissions and performance laws and standards.

3.5.1 Preconstruction Air Permits

Notify the Air Program Manager, through the Contracting Officer, at least 6 months prior to bringing equipment, assembled or unassembled, onto the Installation, so that air permits can be secured. Necessary permitting time must be considered regarding construction activities. Clean Air Act (CAA) permits must be obtained prior to bringing equipment, assembled or unassembled, onto the Installation.

The contractor is responsible for the preparation and submittal of all applicable PREQB Permits.

3.5.2 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include equipment purpose (water heater, building heat, and process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Provide in accordance with paragraph PRECONSTRUCTION AIR PERMITS.

3.5.3 Burning

Burning is prohibited on the Government premises.

3.5.4 Class I [and II] ODS Prohibition

Class I and II ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant.

3.5.5 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer and the DPW Environmental Division.

3.5.6 EPA Certification Requirements

Heating and air conditioning technicians must be certified through an EPA-approved program. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by environmental law.

3.5.7 Dust Control

Always keep dust down, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

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3.5.7.1 Haul Route

Submit truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

3.5.7.2 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be always controlled, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with state and local visibility regulations.

3.5.7.3 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with 29 CFR 1910.

3.5.8 Odors

Odors from construction activities shall be controlled at all times. The odors shall not cause a health hazard and shall be in compliance with State regulations and/or local ordinances.

3.6 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

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Whenever technically and economically feasible the Contractor shall promote deconstruction, salvage, and reuse of building materials and shall limit demolition activities only to those materials permanently affixed to a structure which cannot be disassembled like concrete walls, gypsum board walls, stucco, and brick wall assemblies.

3.6.1 Salvage, Reuse and Recycle

The Contractor shall participate in State and local government sponsored recycling programs. The Contractor shall minimize solid waste generation throughout the duration of the project. Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale, or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

The Contractor shall submit a recycling form stating the amount of material in either pounds or tonnage diverted by its type. The form shall specify amount and type of ferrous or non-ferrous metals, concrete debris, wood (doors, windows, cabinets, etc.) asphalt, etc. to properly sub classify diverted materials. Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

3.6.2 Nonhazardous Solid Waste Diversion Report

The Contractor shall maintain an inventory of non-hazardous solid waste diversion and disposal of construction and demolition debris. The Contractor

shall submit a report to the Contracting Officer in accordance with Section 01110, including non-hazardous demolition debris quantified in cubic yards and tons or pounds. Submit the report to DPW Environmental Office through the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	[_____] tons
Non-Hazardous Solid Waste generated	[_____] tons
Non-Hazardous Solid Waste Recycled	[_____] tons
C&D Debris Recycled	[_____] tons
Total C&D Debris Generated	[_____] tons
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	[_____] tons

Diversion Solid Waste / Construction Debris. Solid Waste Diversion as per Executive Order and the Fort Buchanan Solid Waste Management Plan the contractor is responsible to make the adequate diversion for at least 50% of non-hazardous solid waste from disposal of the construction SOW activities and no less than 75% diversion of construction and demolition materials and debris for disposal.

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3.7 CHEMICAL MATERIALS MANAGEMENT, WASTE MANAGEMENT AND DISPOSAL

3.7.1 Chemicals and Chemical Wastes

The Contractor shall use and store all materials, chemicals, and equipment used in the performance of this contract with these applicable laws. The contract shall follow the Globally Harmonized System of Classification and Labeling of Chemicals (includes SDS requirements). The Contractor shall make all environmental and hazardous material records available, at the beginning of the next duty day after the Government’s request to view the records.

Chemicals shall be dispensed ensuring no spillage to the ground or water. Periodic inspections of dispensing areas to identify leakage and initiate corrective action shall be performed and documented. This documentation will be periodically reviewed by the Government. Chemical waste shall be collected in corrosion resistant, compatible containers. Collection drums shall be monitored and removed to a staging or storage area when contents are within 6 inches of the top. Wastes shall be classified, managed, stored, and disposed of in accordance with Federal, State, and local laws and regulations.

3.7.2 Waste Determination Documentation

Report in memorandum form any Waste Determination for Contractor-derived wastes to be generated. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g., scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of 40 CFR 261 or corresponding applicable state or

local regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination memorandum. As a minimum, provide a Waste Determination memorandum for the following waste (this listing is not inclusive): oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.7.2.1 Sampling and Analysis of Waste

3.7.2.1.1 Waste Sampling

Sample waste in accordance with EPA SW-846. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.

3.7.2.1.2 Laboratory Analysis

Follow the analytical procedure and methods in accordance with the 40 CFR 261. Provide analytical results and reports performed to the Contracting Officer.

3.7.2.1.3 Analysis Type

Identify hazardous waste by analyzing for the following characteristics: ignitability, corrosivity, reactivity, toxicity based on TCLP results.

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3.7.3 Solid Waste Management

3.7.3.1 Solid Waste Management Report

Provide copies of the waste handling facilities, weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation (Contractor certification) must include the receiver's tax identification number and business, EPA, or state registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.7.3.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with 40 CFR 260, state, and local requirements for solid waste disposal. A Subtitle D RCRA permitted landfill is the minimum acceptable offsite solid waste disposal option. Verify that the

selected transporters and disposal facilities have the necessary permits and licenses to operate. Segregate and separate treated wood components disposed at a lined landfill approved to accept this waste in accordance with local and state regulations. Solid waste disposal offsite must comply with most stringent local, state, and federal requirements, including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with 49 CFR 173.

3.7.4 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

3.7.4.1 Hazardous Waste/Debris Management

Work done under this contract is to be performed on a government facility. According to rules and procedures of the United States Environmental Protection Agency, the federal facility is required to have a generator identification number under the Resource Conservation and Recovery Act (RCRA) and to be responsible for wastes (as defined under RCRA) produced, managed, stored, disposed on, or transported from the facility. Accordingly, Contractor will, to the greatest extent practicable, use materials, processes, and techniques that will avoid the creation of hazardous waste.

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a. Contractor shall identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with federal, state, and local regulations, including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, and 40 CFR 268.

b. Contractor shall prepare and follow a written waste management and disposal plan for all hazardous wastes generated on the site. Prior to generation of any hazardous wastes, contractor will coordinate planned activities regarding hazardous materials and hazardous waste with the Contracting Officer.

c. Contractor shall submit a written waste management plan, through the contracting officer, to installation environmental office. Contractor shall follow this plan once it has been approved by the contracting officer.

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. The Contractor shall take sufficient measures to prevent spillage of hazardous and toxic materials during dispensing (reference Asbestos Operations and Maintenance Plan (Asbestos) and Lead Operations and Maintenance Plan (Lead) for the general management and maintenance operations that involve materials containing these toxic substances, excluding abatement). The Contractor shall segregate hazardous waste from other materials and wastes, shall protect it from the weather by placing it in a safe covered location, and shall take precautionary measures against accidental spillage. The Contractor shall be responsible for

storage, describing, packaging, labeling, marking, and placarding of hazardous waste and hazardous material in accordance with 49 CFR 171 - 178, State, and local laws and regulations.

Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government. Prior to removal of any hazardous waste from Government property, hazardous waste manifests must be signed by personnel from the Installation Environmental Office. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in 40 CFR 372-SUBPART D.

3.7.4.2 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with 40 CFR 262.34 and applicable state or local regulations. Individual waste streams will be limited to 55 gallons of accumulation (or 1 quart for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90-day accumulation point be established. Submit a request in writing to the Contracting Officer and provide the following information (Attach Site Plan to the Request):

Contract Number	[_____]
Contractor	[_____]
Haz/Waste or Regulated Waste POC	[_____]
Phone Number	[_____]
Type of Waste	[_____]
Source of Waste	[_____]
Emergency POC	[_____]
Phone Number	[_____]
Location of the Site	[_____]

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Attach a Waste Determination memorandum for the expected waste streams. Allow 10 working days for processing this request. Additional compliance requirements (e.g., training, and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.7.4.3 Hazardous Waste Disposal

3.7.4.3.1 Responsibilities for Contractor's Disposal

The disposition of Contractor generated hazardous waste and excess hazardous materials are the Contractor's responsibility. Provide hazardous waste manifest to the Installations Environmental Office for review, approval, and signature prior to shipping waste off Government property.

3.7.4.3.1.1 Services

Provide service necessary for the final treatment or disposal of the hazardous material or waste in accordance with 40 CFR 260, local, and state, laws and regulations, and the terms and conditions of the Contract within 60 days after the materials have been generated. These services include necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal or transportation, include manifesting or complete waste profile sheets, equipment, and compile documentation).

3.7.4.3.1.2 Samples

Obtain a representative sample of the material generated for each job done to provide waste stream determination.

3.7.4.3.1.3 Analysis

Analyze each sample taken and provide analytical results to the Contracting Officer. See paragraph WASTE DETERMINATION DOCUMENTATION.

3.7.4.3.1.4 Labeling

Determine the Department of Transportation's (DOT's) proper shipping names for waste (each container requiring disposal) and demonstrate to the Contracting Officer how this determination is documented and supported by the sampling and analysis requirements contained herein. Label all containers of hazardous waste with the words "Hazardous Waste" or other words to describe the contents of the container in accordance with 40 CFR 262.31 and applicable state or local regulations.

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3.7.4.3.1.5 Contractor Disposal Turn-In Requirements

Hazardous waste generated must be disposed of in accordance with the applicable Federal, State, and Local regulations including the following conditions to meet installation requirements:

a. Drums must be compatible with waste contents and drums must meet DOT requirements for 49 CFR 173 for transportation of materials.

b. Provide label in accordance with 49 CFR 172.101.

c. Leave 3 to 5 inches of empty space above volume of material.

The Contractor shall transport Contractor generated hazardous waste off Government property immediately or prior to end of work, accordance with the Environmental Protection Agency and the Department of Transportation laws and regulations. The Contractor shall dispose of hazardous waste in compliance with Federal, State, and local laws and regulations, copies of the original and final disposal manifest will also be submitted through the contracting officer, to installation environmental office.

3.7.4.4 Universal Waste Management

Manage the following categories of universal waste in accordance with federal, state, and local requirements and installation instructions:

a. Batteries as described in 40 CFR 273.2

b. Lamps as described in 40 CFR 273.5

c. Mercury-containing equipment as described in 40 CFR 273.4

d. Pesticides as described in 40 CFR 273.3

Mercury is prohibited in the construction of this facility, unless specified otherwise, and except for mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed.

3.7.4.5 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with 40 CFR 260-262, state, and local requirements, and installation instructions.

3.7.4.6 Disposal Documentation for Hazardous and Regulated Waste

Contact the Contracting Officer for the facility RCRA identification number that is to be used on each manifest.

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Submit a copy of the applicable EPA and/or state permit(s), manifest(s), or license(s) for transportation, treatment, storage, and disposal of hazardous and regulated waste by permitted facilities. Hazardous or toxic waste manifests must be reviewed, signed, and approved by the Contracting Officer before the Contractor may ship waste. To obtain specific disposal instructions, coordinate with the Installation Environmental Office. Refer to Section 01 57 19.01 20 SUPPLEMENTAL TEMPORARY ENVIRONMENTAL CONTROLS for the Installation Point of Contact information.

3.7.5 Releases/Spills of Oil and Hazardous Substances

3.7.5.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; Contractor shall immediately (within 15 minutes) notify the Installation Fire Department (787-707-5911), the Installation Command Duty Officer, the Installation Environmental Office, and the Contracting Officer.

Submit verbal and written notifications as required by the federal (40 CFR 300.125 and 40 CFR 355), state, local regulations and as instructed by the Installation Environmental Office and the Contracting Officer instructions. Provide copies of the written notification and documentation that a verbal notification was made within 20 days. Spill response must be in accordance

with 40 CFR 300 and applicable state and local regulations. Contain and clean up these spills without cost to the Government.

Contractor must provide and maintain spill kits onsite ready to use in an emergency.

3.7.5.2 Clean Up

Clean up and cleanup cost of hazardous and non-hazardous waste spills shall be the Contractor's responsibility. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor- responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate, and the spill remains a threat to human health or the environment.

3.7.6 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

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3.7.7 Wastewater

3.7.7.1 Disposal of Wastewater must be as specified below

3.7.7.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related wastewater off-Government property in accordance with 40 CFR 403, state, regional, and local laws, and regulations. Contractor is responsible to characterize as hazardous or non-hazardous any soil or wastewater that would be taken off - Government property as waste and coordinate with an approved landfill for disposition. All waste manifests shall be reviewed and signed by qualified USA Corps of Engineers representative or Government DPW Environmental Officer through the Contracting Officer.

3.7.7.1.2 Surface Discharge

For discharge of ground water, the Contractor shall comply with all Federal and State and Local Regulations; including in accordance with the requirements of the NPDES or state STORMWATER DISCHARGES FROM CONSTRUCTION SITES permit.

3.7.7.1.3 Water Flushing

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing must be discharged into the sanitary sewer in accordance and approval of federal, state, and local laws and regulations.

3.8 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this contract. Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS, and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with 40 CFR 261.

3.9 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds. Consult with the U.S. Department of Agriculture jurisdictional office for additional cleaning requirements.

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3.10 CONTROL AND MANAGEMENT OF ASBESTOS CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos containing waste in accordance with 40 CFR 61. Refer to Section 02 82 13.00 10 ASBESTOS ABATEMENT, and Section 02 82 16.00 20 ENGINEERING CONTROL OF ASBESTOS CONTAINING MATERIALS. Manifest asbestos-containing waste and provide the manifest to the Contracting Officer. Notifications to the state and Installation Air Program Manager are required before starting any asbestos work. Asbestos Abatement Permit it require under Natural Resources and Environmental Department former (Puerto Rico Environmental Quality Board).

3.11 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with 40 CFR 745 and Section 02 82 33.13 20 REMOVAL/CONTROL AND DISPOSAL OF PAINT WITH LEAD; Section 02 83 13.00 20 LEAD IN CONSTRUCTION; and Section 02 83 19.13 10 LEAD-BASED PAINT ABATEMENT. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer. Lead Abatement Permit it require under Natural Resources and Environmental Department former (Puerto Rico Environmental Quality Board).

3.12 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)

Manage and dispose of PCB-contaminated waste in accordance with 40 CFR 761 and Section 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBS).

3.13 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with 40 CFR 761. Refer to Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBS AND MERCURY.

3.14 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered, or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

3.15 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with EPA 40 CFR 112, and other federal, state, regional, and local laws, and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of the United States. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. Used lubricants and used oil to be discarded shall be stored in marked, corrosion-resistant containers and recycled or disposed in accordance with 40 CFR 279, State, and local laws and regulations. [There shall be no storage of fuel on the project site. Fuel must be brought to the project site each day that work is performed. Storage of fuel on the project site must be in accordance with EPA, state, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS.] Bulk storage of petroleum, oil, and lubricants (POL) is prohibited.

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3.15.1 Used Oil Management

Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.15.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 5 inches freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 1320 gallons will be used onsite (only containers with a capacity of 55 gallons or greater are counted), provide and implement a SPCC plan meeting the

requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.16 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer and the DPW Environmental Division at 787-707-3575, 3522, or 3572. Do not disturb this material until authorized by the Contracting Officer.

3.17 PEST MANAGEMENT

To minimize impacts to existing fauna and flora, coordinate with the Installation Pest Management Coordinator (IPMC), through the Contracting Officer, at the earliest possible time prior to pesticide application. Discuss integrated pest management strategies with the IPMC and receive concurrence from the IPMC through the Contracting Officer prior to the application of any pesticide associated with these specifications. Provide Installation Project Office Pest Management personnel the opportunity to be present at meetings concerning treatment measures for pest or disease control and during application of pesticide. For termiticide requirements, see Section 31 31 16.13 CHEMICAL TERMITE CONTROL and Section 31 31 16.19 TERMITE CONTROL BARRIERS. The use and management of pesticides are regulated under 40 CFR 152 - 186.

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3.17.1 Application

Once authorized by the COR and the IPMC apply only DOD Approved pesticides using a state-certified pesticide applicator in accordance with EPA label restrictions and recommendation. The certified applicator must wear clothing and personal protective equipment as specified on the pesticide label. The Contracting Officer will designate locations for water used in formulating. Do not allow the equipment to overflow. Inspect equipment for leaks, clogging, wear, or damage and repair prior to application of pesticide.

3.17.2 Pesticide Treatment Plan

Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application and calibration of equipment. Comply with 40 CFR 152-189, state, regional, and local pest management record-keeping and reporting requirements as well as any additional Installation Project Office specific requirements in conformance with [DA AR 200-1 Chapter 5, Pest Management, Section 5-4 "Program requirements"] for data required to be reported to the Installation.

3.18 SOUND INTRUSION

The Contractor shall keep construction activities under surveillance and control to minimize environment damage by noise. The Contractor shall comply with Federal and State regulations. Make the maximum use of low-noise emission products, as certified by the EPA. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and then only during the designated times. Confine pile-driving operations to the period between [7 a.m.] and [4 p.m.], Monday through Friday, exclusive of holidays, unless otherwise specified.

Keep construction activities under surveillance and control to minimize environment damage by noise. Comply with the provisions of the State and local (Commonwealth of Puerto Rico) Regulations (PREQB-Control of Noise Pollution Regulations).

3.19 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

3.20 SPECIAL EXCAVATION CONSIDERATIONS

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Before any excavation is performed within the installation boundaries, the contractor must fill and submit to the DPW engineering office a completed excavation permit for "Request of Excavation/Borings Permit in Fort Buchanan" application for review and approval. The DPW environmental division will evaluate the proposed excavation area if it will be located within environmental restoration sites, munitions sites, and compliance restoration sites, cultural resource sites and endangered species sites. The DPW environmental office will provide a munitions explosives of concern (MEC) awareness pamphlet to every excavation permit application. It is the responsibility of the contractor to follow the protocol of the 3r's (recognize, retrieve and report) if a MEC is encountered.

If the proposed excavation is located within the Munitions Response Site (MRS) identified as the Former Camp Buchanan Training Area(Western half of the golf course, Maxi Williams Jr. baseball field; 1300 Building Area, Mogote Area); Before any proposed construction or excavation work can begin, the work needs to be coordinated with the Department of Public Works at Fort Buchanan and obtain: approval from the Master Planner at Fort Buchanan, receive construction support materials from the Environmental Division, and an approved excavation permit from the DPW Engineering office. Contractor shall screen any debris for munitions if transported outside Fort Buchanan.

3.20.1 Munitions Excavation of Concern Safety Guidance for Work Conducted at Fort Buchanan

a. Contractor is responsible for ensuring that all its employees, sub-contractors or agents working at the site have received the Munitions Explosive of Concern (MEC) Awareness information consisting of:

1. Discussion and Understanding of Sergeant Woof Poster's guidelines; and

2. Munitions Explosives of Concern (MEC) Awareness Pamphlet of MEC items found at Fort Buchanan.

b. Contractor is responsible of visibly posting Spanish and English versions of Sergeant Woof posters at the worksite.

c. Contractor is responsible of visibly posting at the worksite a MEC Awareness Pamphlet of MEC items found at Fort Buchanan.

d. Contractor's employees are responsible of adhering to the following protocol upon detection of suspected ammunition:

1. Recognize that you may have found ammunition and do not touch, move, or disturb the suspected ammunition.
2. Retreat at least 300-feet (a cell phone may set it off); and,
3. Report finding to Fort Buchanan Directorate of Emergency Services at 787-707-4911 / 5911 / 3337 / 3410.

e. Contractor is responsible of posting as many observers as it deems necessary during the excavation process and that may assist in recognizing any ammunition present.

1. MEC: This term, which distinguishes specific categories of military munitions that may pose unique explosives safety risks and is identified as:
   - Unexploded Ordnance (UXO), as defined in 10 U.S.C. 2710(e)(9).
   - Discarded Military Munitions (DMM), as defined in 10 U.S.C 2710. (e)(2).
   - Munitions Constituents (MC) (e.g., TNT, RDX) present in high enough concentrations to pose an explosives hazard.

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If the site is located within Restoration or Compliance Restoration Sites, it is the responsibility of the contractor and their Certified Industrial Hygienist (CIH) to protect their employees of contamination by providing the Proper Personnel Protection Equipment (PPE) for the site conditions as identified by the DPW Environmental Office.

The contractor is responsible to characterize (if hazardous or non-hazardous) any soil, sediment or liquid/water if taken offsite as waste and to be disposed according to all Federal, AR Regulation 200-1 and state regulations.

If the site is located within any cultural areas or endangered species habitat an endangered species pamphlet will be provided to the contractor by the DPW Environmental office. The contractor must take all precautionary steps to preserve the Cultural areas and protect endangered species areas. The DPW Engineering Office will receive the excavation permit application from the DPW Environmental Division and the NEC division and proceed with the review process of the excavation permit application for all existing utilities etc. before providing the contactor the final and approved excavation permit application. The contractor cannot start the excavation prior to receiving an approved excavation permit.

It is the responsibility of the contractor to include at a minimum the safety guidance listed above and the protocol of Munitions Explosives Concern (MEC) encounters into their Site Safety Plan. The contractor will follow the 3Rs (Recognize it is a possible munitions, do not touch or move the item at any time, Retreat 300-feet and Report - to Fort Buchanan Security 787-707-4911). The Safety Plan shall be reviewed by Fort Buchanan's Safety Office. The contractor shall include the MEC Awareness topic in their Site Safety program, tailgate safety meetings and require every employee working on site to sign an acknowledgement that they have received the MEC Awareness training. It is the responsibility of the Contracting Office Representative (COR) that the MEC Awareness Materials has been received by the contractor, and MEC information is in the contractor's site safety plan and that the information has been presented to each employee working on site by the contractor.

It is the responsibility of the contractor to coordinate a MEC, Boa and Pollution Prevention awareness training with the Fort Buchanan Environmental Office before any excavation takes place.

-- **End of Section** --

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DEPARTMENT OF THE ARMY
U.S. ARMY INSTALLATION MANAGEMENT COMMAND
HEADQUARTERS, UNITED STATES ARMY GARRISON, FORT BUCHANAN
390 DOUBLE EAGLE AVENUE, SUITE 300
FORT BUCHANAN, PUERTO RICO 00934-4614

AMIM-BCP (1oo) 30 July 2023

MEMORANDUM FOR All personnel assigned to or under operational control of the United States Army Garrison (USAG) Fort Buchanan

SUBJECT: Installation Policy No. 23-17, Fort Buchanan Environmental Sustainability

1. References:

a. National Environmental Policy Act (NEPA), 42 United States Code Sections 4321 et seq.

b. Army Regulation (AR) 200-1, Environmental Protection and Enhancement, 13 December 2007.

c. Army Regulation (AR) 420-1, Army Facilities Management, 12 February 2008, RAR, 24 August 2012.

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d. 32 Code of Federal Regulations Part 651, Environmental Analysis of Army Actions, 29 March 2002.

e. EO 14057: Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability, December 8, 2021

f. EO 14008: Tackling the Climate Crisis at Home and Abroad, January 27, 2021

g. Army Directive 2020-08, U.S. Army Installation Policy to Address Threats Caused by Changing Climate and Extreme Weather, 11 September 2020.

h. Army Climate Strategy Implementation Plan Fiscal Years 2023-2027

2. Purpose: To define Fort Buchanan's Environmental and Sustainability Policy and commitments.

3. Applicability: All Soldiers, Civilian Employees, Residents and Visitors present at Fort Buchanan.

4. Policy: Fort Buchanan will implement programs that will ensure that land, air, and water resources are always available for Soldier readiness. Our Sustainability Programs will prioritize renewable energy and water sources; increasing waste diversion and recycling solutions; and implement Greenhouse Gas emission reduction measures. In keeping with this policy, a sustainable Fort Buchanan will proactively address environmental sustainability by making intelligent use of our natural and technological resources and keeping our workforce and community committed to these goals. Fort Buchanan's priorities will be as directed herein:

a. Compliance: Military, Civilian employees, and contractors, will comply with applicable Federal, State and Local environmental laws. All new activities arising from Garrison or Tenant contracted actions and services must comply with the provisions contained in the pertinent Environmental Protection Scope of Work. If in doubt as to whether a given statute or requirement is applicable or not, a legal opinion from the Garrison's Installation Legal Office will be sought and complied with. Any individual may alert the Garrison Chain of Command through any means if he/she reasonably believes that an environmental violation has taken or will take place. No reprisals will be taken against any such individual. Suspected violations will be investigated as deemed fit and any appropriate corrective course of action will be carried out.

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b. Energy Conservation: Every building occupant within Fort Buchanan, whether military, employee, or visitor, is responsible for eradicating energy waste and misuse. All building occupants will endeavor to achieve the objectives and targets set by Fort Buchanan's Environmental Management Action Plans (EMAPs) for Energy Security and Conservation.

c. Water Consumption: Every tenant and building occupant within Fort Buchanan will consume water in a responsible manner and as if it were she/he that was paying for it. All building occupants will endeavor to achieve the objectives and targets set by Fort Buchanan's EMAP for Water Conservation.

d. Solid Waste and Recycling: Every tenant and building occupant will endeavor to reduce solid waste and recycle as much as reasonably possible. All building occupants will endeavor to achieve the objectives and targets set by Fort Buchanan's EMAP for Solid Waste and Recycling. Mixing recyclable waste with non-recyclable waste is strictly forbidden. The Directorate of Public Works (DPW) Environmental Division is responsible for ensuring that everyone at Fort Buchanan understands our recycling parameters and that the same are strictly enforced.

e. Protecting Natural: Preserving the natural resources will comply with pertinent environmental statutes; to maintain available those assets for future training lands. Tenants must consult with the Environmental Office prior to any potential alterations to its natural.

f. Climate-Change: Greenhouse gas emissions and climate-change effects resulting from proposed Army actions will be assessed and minimized in accordance with existing Army guidance and instructions. The results of climate-change prediction analysis tools will be added into all facilities and infrastructure-related plans, policies, and procedures.

5. Proponent: The proponent of the policy is the Directorate of Public Works (DPW), Environmental Division. The point of contact for this matter is the Fort Buchanan's Environmental Division, telephone (787) 707- 3575. DPW will keep this Command abreast of those tenants and building occupants that are either delinquent in complying with this policy or excel in carrying out its goals.

CHARLES N. MOULTON
COL, LG
Commanding

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